

NINGBO, China

WMO: 582390

Lat: 29.8153N

Lon: 121.4740E

Elev: 4

StdP: 101.28

Time Zone: 8.00 (E08)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-1.0	-11.1	1.5	4.8	-8.2	1.9	4.8	8.3	8.5	7.3	7.6	1.6	310	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	37.1	27.5	35.8	27.3	34.1	27.1	28.5	34.0	28.0	33.0	27.5	32.1	3.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.3	26.9	22.6	30.2	26.2	21.6	29.7	91.5	34.0	89.6	33.6	87.5	32.6	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.9	6.1	-4.5	39.5	1.6	1.1	-5.6	40.3	-6.6	40.9	-7.5	41.5	-8.6	42.3		
			-5.3	29.5	1.8	0.5	-6.6	29.9	-7.7	30.2	-8.7	30.5	-10.0	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	5.8	7.2	11.5	17.1	21.9	24.9	29.9	29.6	25.1	19.9	14.1	7.8	
	DBStd	8.89	3.61	3.76	4.07	3.75	2.89	2.85	2.64	2.21	2.65	2.70	3.80	3.57	
	HDD10.0	359	138	96	30	1	0	0	0	0	0	0	9	86	
	HDD18.3	1459	387	313	214	67	5	0	0	0	0	13	134	325	
	CDD10.0	3267	9	16	77	213	367	446	617	608	454	308	132	19	
	CDD18.3	1324	0	0	3	29	114	196	359	350	204	63	7	0	
	CDH23.3	13092	1	1	28	203	731	1472	4482	4282	1598	267	28	0	
	CDH26.7	5652	0	0	2	42	215	517	2289	2057	486	41	4	0	
Wind	WSAvg	2.8	2.7	2.7	3.0	3.0	2.9	2.5	3.2	3.1	2.6	2.7	2.5	2.7	
Precipitation	PrecAvg	1483	76	77	121	112	128	227	143	208	170	79	81	61	
	PrecMax	1944	217	148	278	248	223	430	335	358	364	274	237	170	
	PrecMin	914	24	16	41	45	49	73	22	57	55	4	21	2	
	PrecStd	222	38	37	65	47	52	85	71	80	73	65	55	46	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	21.6	26.1	30.2	33.2	36.1	39.2	39.0	34.1	30.8	26.8	20.2	
		MCWB	13.4	14.7	17.2	19.6	23.8	27.8	27.5	27.7	26.4	25.5	20.2	15.3	
	2%	DB	16.9	18.8	23.8	27.9	31.1	33.8	37.8	37.0	32.8	27.2	23.8	17.8	
		MCWB	11.9	13.7	16.8	19.2	22.2	27.2	27.6	27.4	25.7	21.5	19.0	13.7	
	5%	DB	13.9	16.0	20.9	25.9	29.1	31.8	36.1	35.2	31.0	25.9	21.2	15.2	
		MCWB	10.2	12.4	14.9	18.7	21.5	25.7	27.5	27.0	25.1	20.0	17.7	11.6	
	10%	DB	11.8	13.8	18.8	23.8	27.1	29.8	34.9	34.0	29.2	24.8	20.0	13.9	
		MCWB	8.9	10.7	14.1	18.1	20.6	25.1	27.3	27.2	24.3	19.6	16.8	10.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	17.0	20.0	23.1	26.1	28.6	29.2	29.1	27.5	26.0	22.7	18.0
			MCDB	17.8	18.6	23.7	26.7	31.8	33.7	35.6	35.0	31.4	28.9	24.8	19.0
2%		WB	13.1	14.8	17.7	21.7	24.5	27.6	28.6	28.5	26.7	24.3	20.8	14.6	
		MCDB	15.4	17.4	21.2	24.8	28.1	31.6	34.5	33.8	29.9	25.6	22.1	16.1	
5%		WB	11.2	13.1	16.2	20.5	23.6	26.7	28.1	28.0	26.2	22.5	19.2	13.1	
		MCDB	13.5	15.6	19.9	23.4	26.7	30.3	33.6	32.7	29.1	24.4	20.4	14.7	
10%		WB	9.2	11.0	14.7	19.3	22.6	25.9	27.6	27.5	25.6	21.1	17.8	11.2	
		MCDB	10.9	13.2	17.7	22.3	25.1	28.6	32.7	31.8	28.3	23.2	19.3	12.7	
Mean Daily Temperature Range		5% DB	MDBR	7.2	7.4	8.8	9.6	8.7	6.6	8.3	7.7	7.0	8.0	7.7	8.1
			MCDBR	12.5	13.0	13.4	13.1	12.0	9.9	10.6	10.0	8.9	9.8	10.6	11.5
	MCWBR		7.9	7.9	7.3	6.2	5.1	3.9	3.3	3.0	3.3	4.4	5.6	6.9	
	5% WB	MCDBR	10.3	10.5	11.2	9.6	8.5	8.3	9.6	8.5	7.5	6.4	6.8	7.9	
		MCWBR	7.2	7.4	7.2	5.6	4.6	3.8	3.4	3.0	3.4	4.0	4.6	6.0	
Clear-Sky Solar Irradiance	taub	0.614	0.656	0.699	0.708	0.675	0.693	0.645	0.658	0.632	0.610	0.605	0.590		
	taud	1.631	1.578	1.532	1.565	1.666	1.674	1.795	1.752	1.776	1.736	1.702	1.683		
	Ebn at Noon	592	617	630	649	674	660	690	675	672	645	596	584		
	Edh at Noon	217	249	276	276	250	247	218	226	214	209	200	197		
All-Sky Solar Radiation	RadAvg	2.22	2.58	3.54	4.22	4.50	4.04	5.61	4.91	3.82	3.39	2.43	2.25		
	RadStd	0.35	0.57	0.32	0.49	0.56	0.58	0.62	0.56	0.46	0.51	0.34	0.40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+79	+51

Nomenclature: See separate page